Lab 17 — Site-to-Site VPN with WireGuard

You will build an encrypted tunnel between two simulated sites running in Linux network namespaces.

Run on https://killercoda.com/playgrounds/scenario/ubuntu

Required software (free, apt): - `wireguard`, `wireguard-tools` - `iproute2` (pre-installed)

Step 1 — Install WireGuard

```
apt update && apt install -y wireguard wireguard-tools
```

```
modprobe wireguard
```

```
lsmod | grep wireguard
```

Step 2 — Generate two key pairs

```
mkdir -p ~/wg && cd ~/wg
```

```
wg genkey | tee siteA.key | wg pubkey > siteA.pub
```

```
wg genkey | tee siteB.key | wg pubkey > siteB.pub
```

Step 3 — Build two namespaces (Site A, Site B)

```
ip netns add siteA

ip netns add siteB

ip link add wgA type wireguard

ip link add wgB type wireguard

ip link set wgA netns siteA

ip link set wgB netns siteB
```

Step 4 — Configure each tunnel endpoint

```
ip netns exec siteA wg set wgA private-key siteA.key listen-port
51820 \
```

```
    peer "$(cat siteB.pub)" allowed-ips 10.99.0.2/32

ip netns exec siteB wg set wgB private-key siteB.key listen-port
51821 \

    peer "$(cat siteA.pub)" allowed-ips 10.99.0.1/32

ip netns exec siteA ip addr add 10.99.0.1/24 dev wgA

ip netns exec siteB ip addr add 10.99.0.2/24 dev wgB

ip netns exec siteA ip link set wgA up

ip netns exec siteB ip link set wgB up
```

Step 5 — Connect the tunnels via host loopback

```
ip netns exec siteA wg set wgA peer "$(cat siteB.pub)" endpoint
127.0.0.1:51821
```

```
ip netns exec siteB wg set wgB peer "$(cat siteA.pub)" endpoint 127.0.0.1:51820
```

Step 6 — Verify

```
ip netns exec siteA ping -c3 10.99.0.2

ip netns exec siteA wg show

ip netns exec siteB wg show
```

`latest handshake` and incrementing `transfer` bytes prove the tunnel is up.

Step 7 — Compare with IPsec / OpenVPN

Property	WireGuard	IPsec	OpenVPN
Lines of kernel code	~4000	~400000	user space
Cryptoagility	None — fixed suite	Negotiated	Negotiated
Roaming clients	Built-in	Add-on (MOBIKE)	Built-in

WireGuard's small surface area is itself a security control.

Free online tools

- **Tailscale (free tier, mesh on top of WireGuard)** — https://tailscale.com
- **NetMaker** — https://www.netmaker.io
- **OpenVPN Community** — https://openvpn.net/community/
- **strongSwan IPsec** — https://www.strongswan.org
- **WireGuard config generator** — https://www.wireguardconfig.com

---

What you learned

- The four artefacts a WireGuard tunnel needs (priv key, peer pub key, endpoint, allowed-ips).
- Why "allowed-ips" doubles as routing **and** crypto-key routing.
- How a small protocol surface translates to a smaller attack surface.